



Analysis and Forecasting of Syphilis Trends in Mexico

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Econometrics 1
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Introduction

With the objective of helping awareness and research to mitigate syphilis in Mexico, two time series models were developed: SARIMA and LSTM. In which we seek to obtain a model that can accurately predict the trend of syphilis in Mexico. This idea arose from the article referenced in [1]. Syphilis infection data spans from January 2004 to December 2023. For the training of the models, the first 228 months were used, leaving the last 12 months as test data.

Methods

SARIMA Model

The Seasonal Autoregressive Integrated Moving Average model is a statistical method that extends the ARIMA model to account for seasonal patterns, it is represented as (p, d, q)(P, D, Q)s. “d” and “D” represent the degree of nonseasonal and seasonal differencing, respectively. “q” and “Q” represent the number of lags in the non-seasonal and seasonal moving average components, respectively. “s” represents the periodicity of the time series, i.e. how often seasonal patterns repeat (e.g. 12 for monthly data).

The SARIMA model can be expressed in the following polynomial form:

$$\begin{aligned}\phi(L)\Phi(L)\Delta^d\Delta_s^D y_t &= \theta(L)\Theta(L)\varepsilon_t, \\ L^i y_t &= y_{t-i}, \\ \Delta^d &= (1-L)^d, \\ \Delta_s^D &= (1-L^s)^D, \\ \phi(L) &= 1 - \phi_1 L - \dots - \phi_p L^p, \\ \Phi(L) &= 1 - \Phi_1 L - \dots - \Phi_P L^P, \\ \theta(L) &= 1 + \theta_1 L + \dots + \theta_q L^q, \\ \Theta(L) &= 1 + \Theta_1 L + \dots + \Theta_Q L^Q\end{aligned}$$

LSTM Model

The LSTM (Long Short-Term Memory) network is a type of RNN (Recurrent Neural Network) that decides which information to further process and which to leave aside (delete) as information passes through it at each step. This network consists of a sequence input layer, an LSTM layer, and an output layer. The input layer receives time series (TS) data, while the LSTM layer understands the long-term dependencies of the data. Unlike RNNs, LSTM is designed to address the vanishing gradient problem, which refers to the difficulty RNNs have in propagating the error signal, making it difficult for them to learn long-term patterns. Mathematical equations:

$$\begin{aligned}f_t &= \sigma(W_f \cdot [S_{t-1}, X_t] + b_f) \\ i_t &= \sigma(W_i \cdot [S_{t-1}, X_t] + b_i) \\ g_t &= \tanh(W_g \cdot [S_{t-1}, X_t] + b_g) \\ o_t &= \sigma(W_o \cdot [S_{t-1}, X_t] + b_o) \\ C_t &= f_t \odot C_{t-1} + i_t \odot g_t \\ S_t &= o_t \odot \tanh(C_t)\end{aligned}$$

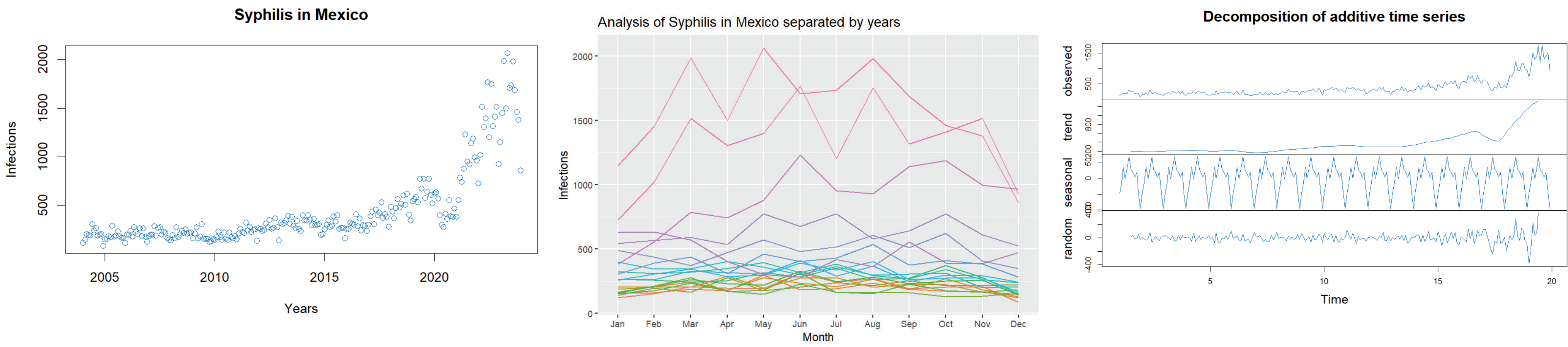
Conclusion

The SARIMA model shows good fit in the training data, but its predictive ability is limited. This is because the residuals of the model do not follow a normal distribution, which affects the validity of the predictions, may be due to the model not being able to capture the underlying factors to the evolution of syphilis. From this we conclude that the SARIMA model is not efficient to predict future data. Given the good metrics of the LSTM model, this model by itself is sufficient to make predictions of the number of syphilis infections in Mexico; it is only necessary to correct the method to properly detransform the data. Despite the limitations of the model, the trend observed in the transformed data suggests that syphilis will continue to grow if new, more effective prevention and control measures are not implemented. It is crucial that public health authorities strengthen early detection, treatment and education programs to stop the spread of the disease.

References

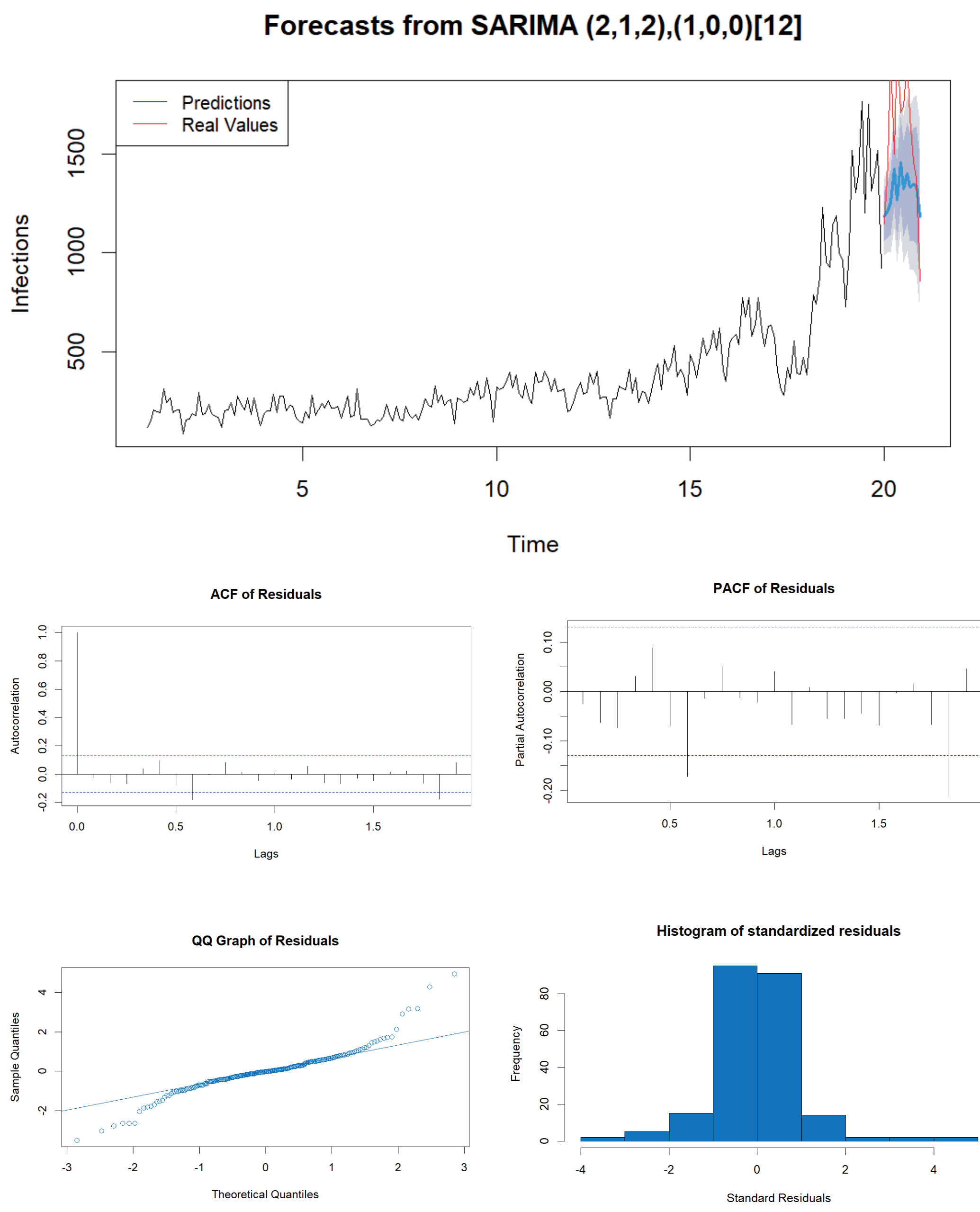
[1]Wang, Z. D., Yang, C. X., Zhang, S. K., Wang, Y. B., Xu, Z., & Feng, Z. J. (2024). Analysis and forecasting of syphilis trends in mainland China based on hybrid time series models. Epidemiology & Infection, 152.
[2] François Chollet. Deep Learning with Python. Manning Publications Co., 2018.
[3] Secretaría de Salud. Anuario de Morbilidad 1984 - 2023. 2023. url: https://epidemiologia.salud.gob.mx/anuario/html/casos_mes.html

Preliminary analysis of the syphilis situation in Mexico

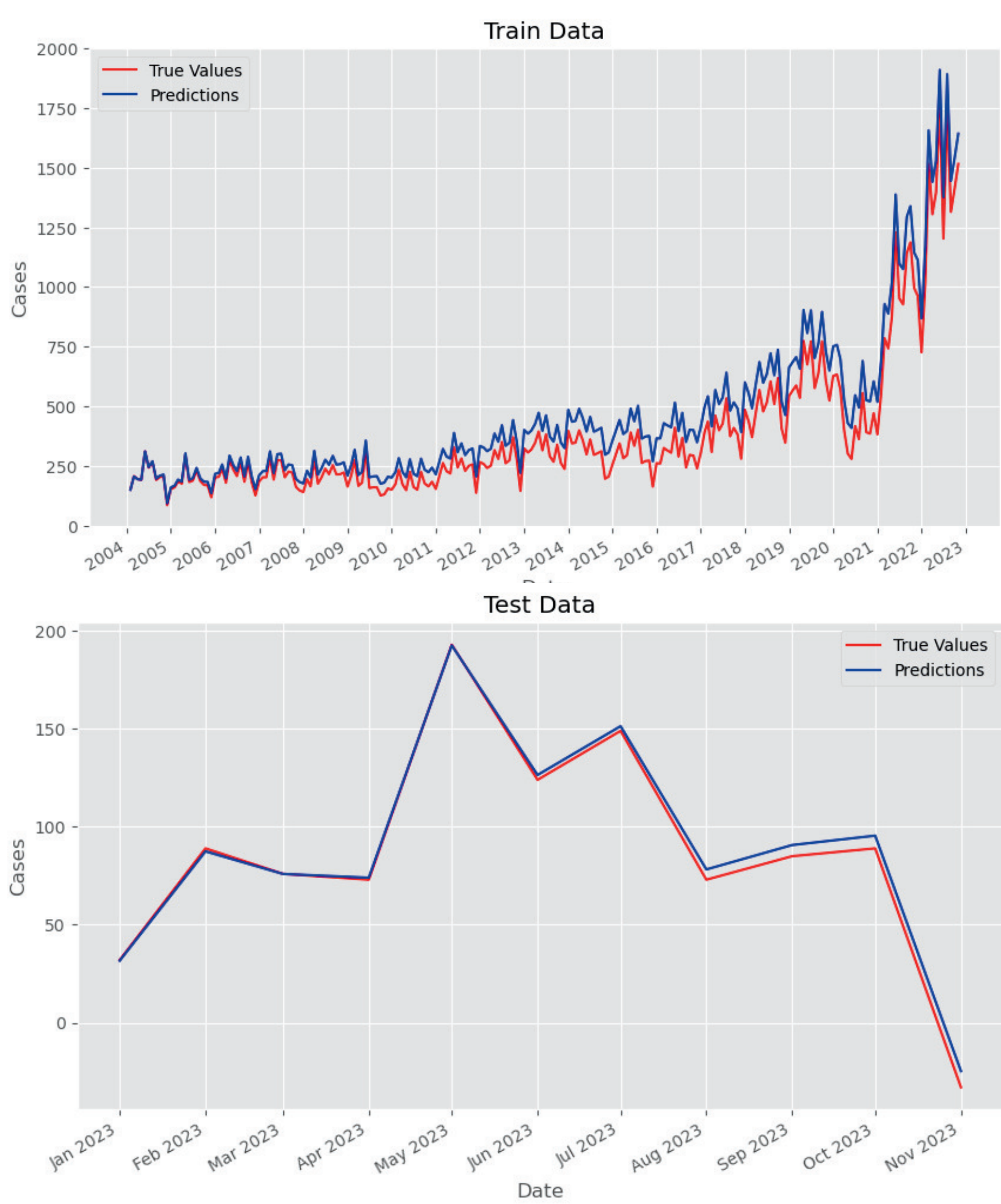


Results

SARIMA Model



LSTM Model



Goodness-of-fit checks of models

Model	Fitting					Testing				
	R^2	MAD	MAPE	RMSE	MAE	R^2	MAD	MAPE	RMSE	MAE
SARIMA	89.73%	42.15	17.61%	97.16	63.87	19.03%	287.60	19.76%	413.3	329.37
LSTM	91.10%	78.44	22.78%	90.00	78.44	99.46%	3.08	4.93%	4.1	3.08